Linux

Server

Troubleshooting:

A Practical Guide

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Server is slow

1. Server is slow. How do you troubleshoot?

First I check **system resource usage**.

Check CPU usage

```
top
```

Check memory usage

```
free -m
```

Check disk space

```
df -h
```

Check I/O wait

```
iostat -x 1
```

Check high CPU processes

```
ps -eo pid,ppid,cmd,%mem,%cpu --sort=-%cpu | head
```

If any process is consuming high CPU or memory, I investigate that process.

2. Disk is full. What will you do?

First I check disk usage.

```
df -h
```

Then I find large files or directories.

```
du -ah / | sort -rh | head -20
```

Then I clean unnecessary logs or files.

```
journalctl --vacuum-time=7d
```

Finally I remove unused files or rotate logs.

3. Service is not starting. How do you troubleshoot?

First I check service status.

```
systemctl status <service-name>
```

Then I check logs.

```
journalctl -xe
```

If required I restart the service.

```
systemctl restart <service-name>
```

4. How do you check open ports?

```
netstat -tulnp
```

or

```
ss -tulnp
```

This shows listening ports and associated processes.

5. How do you check memory leak?

First check overall memory.

```
free -m
```

Then check processes consuming high memory.

```
top
```

or

```
ps aux --sort=-%mem | head
```

6. How do you check if a server is reachable?

First check network connectivity.

```
ping <ip>
```

Then check port connectivity.

```
telnet <ip> <port>
```

or

```
nc -zv <ip> <port>
```

7. How do you check CPU usage history?

```
sar -u 1 5
```

It shows CPU usage statistics.

8. What will you do if the system hangs?

First check system load.

```
uptime
```

Then check running processes.

```
top
```

Check kernel messages.

```
dmesg | tail
```

9. How do you find a zombie process?

```
ps aux | grep Z
```

10. How do you check file permissions?

```
ls -l
```

This shows read, write, and execute permissions for user, group, and others.

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